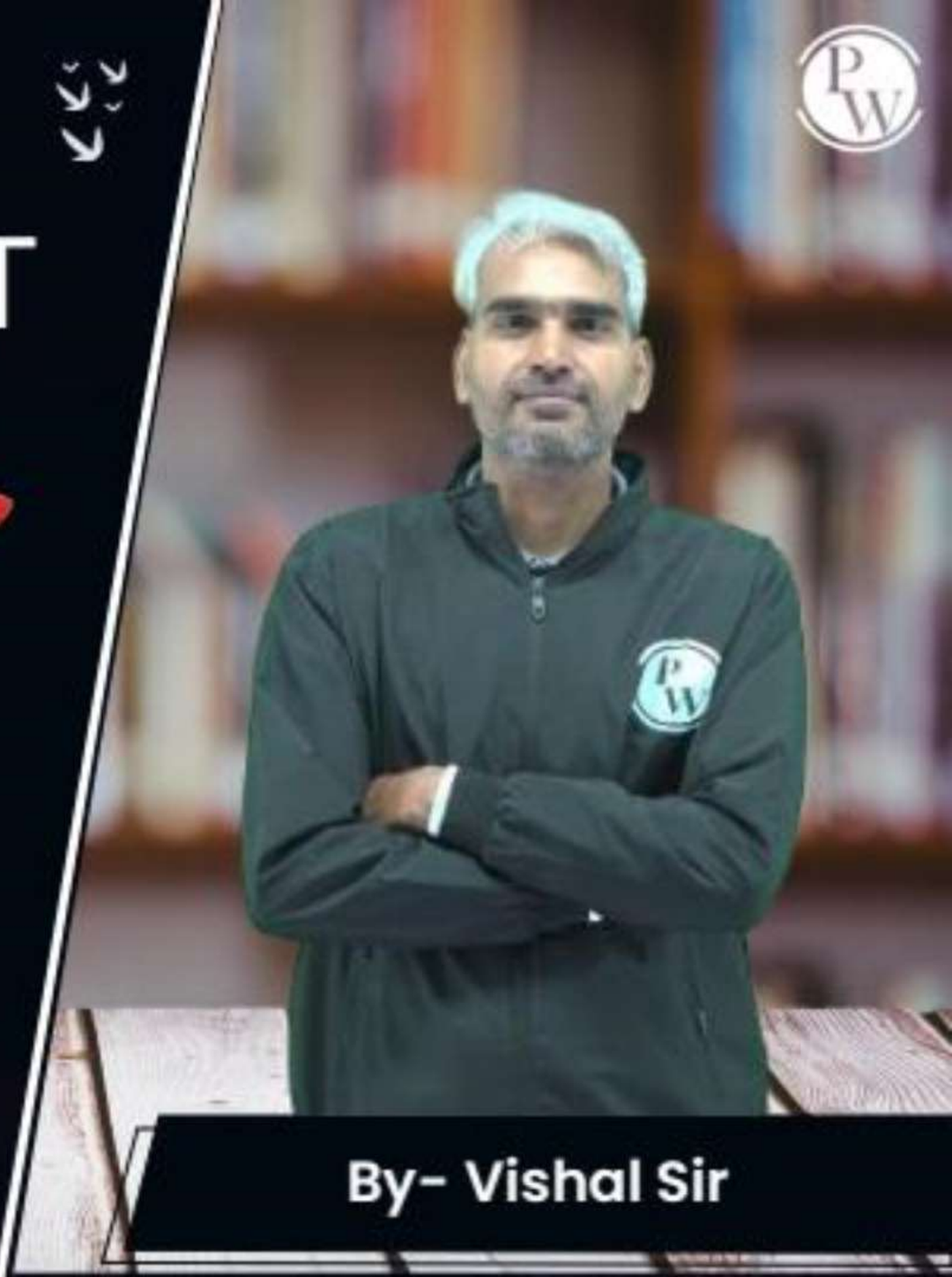
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



Topic Schema refinement (Normalization)

Topic Problems because of redundancy in a relation

Slide

Topics to be Covered



Topic Properties of decomposition

Topic Dependency preserving decomposition

Slide



Topic : Properties of decomposition

Properties of Decomposition

Dependency Preserving decomposition

let Relation R with FD set F is decomposed into two relations R_1 & R_2

R_1 {FD set: F_1} R_2 {FD set: F_2}

In general, $F_1 \cup F_2 \subseteq F$

(i) if $F_1 \cup F_2 = F$, then dep. preserving decomposition

(ii) if $F_1 \cup F_2 \subset F$, not dep. preserving

(iii) $F_1 \cup F_2 \supset F$ {not possible}

Lossless Join Decomposition

let Relation R with FD set F is decomposed into R_1 & R_2

R_1 (FD set = F_1) — Natural Join — R_2 (FD set = F_2)

In general $R_1 \bowtie R_2 \supseteq R$

(i) $R_1 \bowtie R_2 = R$, then lossless Join decomposition

(ii) $R_1 \bowtie R_2 \supset R$, then lossy Join decomposition

(iii) $R_1 \bowtie R_2 \subset R$, it is not possible



Topic : Dependency preserving decomposition

* Let relation R with FD set F is decomposed into sub-relations $R_1, R_2, R_3, \ldots R_m$ with FD sets $F_1, F_2, F_3, \ldots F_m$ respectively,

In general, $(F_1 \cup F_2 \cup F_3 \ldots \cup F_m) \subseteq F$

① if $=$ then dep. preserving

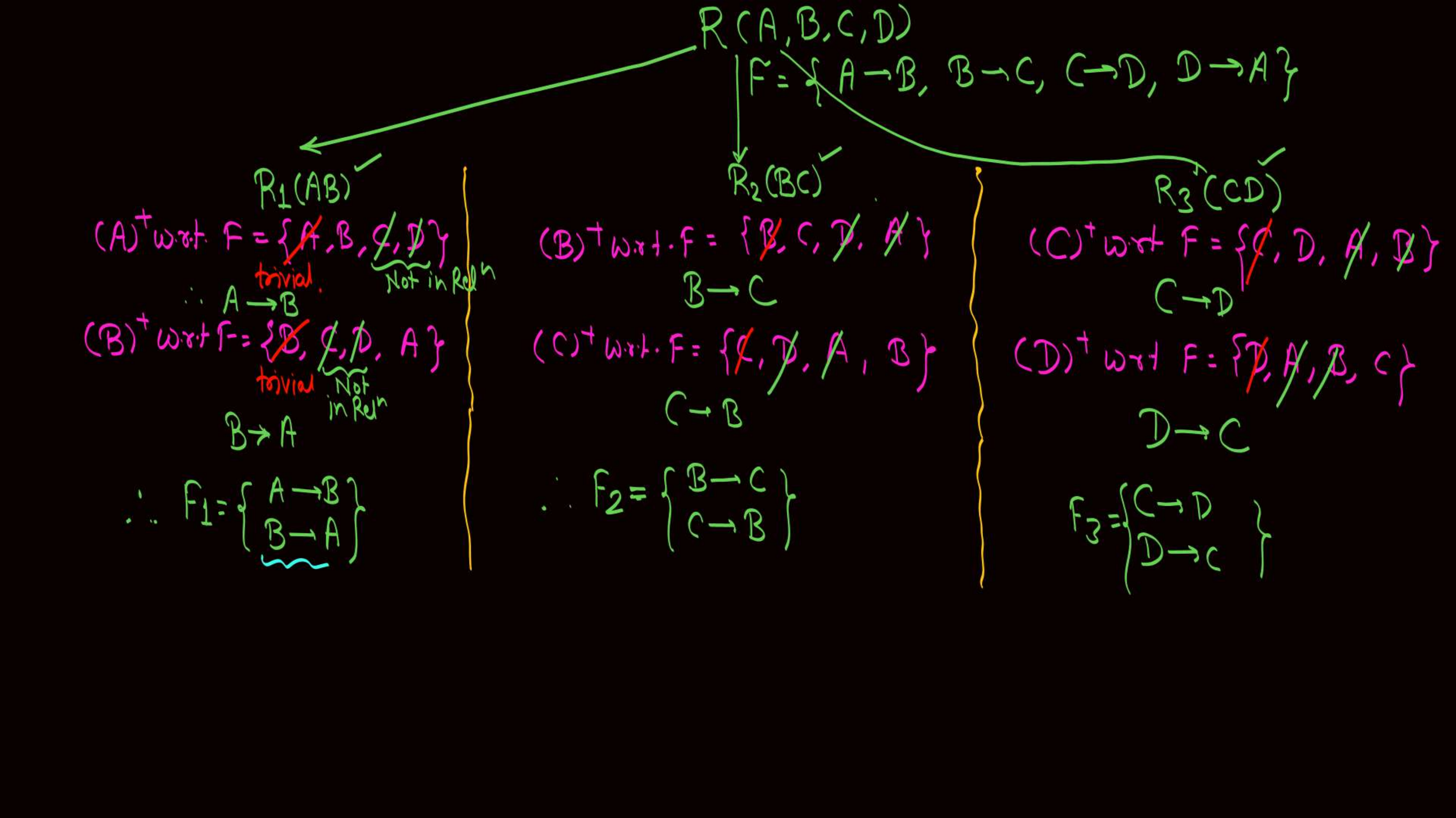
② if $\subset$ then not dep. preserving



Topic : Dependency preserving decomposition

Q: Let $R(A, B, C, D)$ be the relational schema with FD set $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

If relation R is decomposed into three sub-relations $R_1(AB)$, $R_2(BC)$, $R_3(CD)$ then, check whether this decomposition is dependency preserving decomposition or not?


R(A, B, C, D)
F = { A → B, B → C, C → D, D → A }
R1(AB)
(A)+ wrt F = { A, B, C, D }
trivial
Not in Rel^n
∴ A → B
(B)+ wrt F = { B, C, D, A }
trivial
Not in Rel^n
B → A
∴ F1 = { A → B, B → A }
R2(BC)
(B)+ wrt F = { B, C, D, A }
B → C
(C)+ wrt F = { C, D, A, B }
C → B
∴ F2 = { B → C, C → B }
R3(CD)
(C)+ wrt F = { C, D, A, B }
C → D
(D)+ wrt F = { D, A, B, C }
D → C
F3 = { C → D, D → C }

$$F_1 \cup F_2 \cup F_3 = \{ A \to B,\ B \to C,\ C \to D,\ B \to A,\ C \to B,\ D \to C \}$$

* We need to check the relationship b/w $(F_1 \cup F_2 \cup F_3)$ & F

① We already know F covers $F_1 \cup F_2 \cup F_3$

i.e $\boxed{F \supseteq F_1 \cup F_2 \cup F_3}$ —— eq^n ①

② Check if $F_1 \cup F_2 \cup F_3$ covers F or not

FDs of F

$A \to B$ {belongs to $F_1 \cup F_2 \cup F_3$} $C \to D$ { belongs to $F_1 \cup F_2 F_3$ }

$B \to C$ {—//—//—} $D \to A$ $(D)^+$ w.r.t $F_1 \cup F_2 \cup F_3 = \{D, C, B, A\}$

$A \in (D)^+$ w.r.t $F_1 \cup F_2 \cup F_3$ ∴ $D \to A$ is a member of $F_1 \cup F_2 \cup F_3$

{ $F_1 \cup F_2 \cup F_3$ Covers F

$F_1 \cup F_2 \cup F_3 \supseteq F$

eq^n — ②

By eqn ① & ②

$$F_1 \cup F_2 \cup F_3 = F$$

∴ decomposition is dependency Poeserving decomposition

Q: Let R(A B C D) be the relational schema with FD set $F = \{AB \to CD, D \to A\}$

If relation R is decomposed as $D = \{R_1(B C D), R_2(A D)\}$

Check whether decomposition D of relation R is dep. preserving or not?

$(B)^+$ w.r.t $F = \{B\}$

$(C)^+$ -,,- $= \{C\}$

$(D)^+$ -,,- $= \{D, A\}$

$(BC)^+$ -,,- $= \{B, C\}$

$(BD)^+$ -,,- $= \{B, D, A, C\} \Rightarrow BD \to C$

$(CD)^+$ -,,- $= \{C, D, A\}$

$(A)^+$ w.r.t $F = \{A\}$

$(D)^+$ w.r.t $F = \{D, A\}$

$D \to A$

$F_1 \cup F_2 = \{BD \to C, D \to A\}$

① $F_1 \cup F_2 \subseteq F$ is always true

② Check if $F \subseteq F_1 \cup F_2$

FD of F

$AB \to CD \Rightarrow (AB)^+$ w.r.t $F_1 \cup F_2 = \{A, B\}$

$CD \notin (AB)^+$ w.r.t. $F_1 \cup F_2$

$\therefore AB \to CD$ is not a member of $F_1 \cup F_2$

$\therefore F \nsubseteq F_1 \cup F_2$

i.e. $F_1 \cup F_2 \neq F$

hence not a dep. preserving decomposition.

Q: Let R(ABCDEF) be the relational schema with FD set $F = \{AB \to C, BC \to A, AC \to B, B \to D, AD \to E, E \to F\}$

Which of the following decomposition of R is dependency preserving decomposition

$D_1 = \{R_1(ABC), R_2(ACDE), R_3(ADF)\}$

$F_1 = \{AB \to C, BC \to A, AC \to B\}$

$F_2 = \{AD \to E, AC \to DE, ACD \to E, ACE \to D\}$

$F_3 = \{AD \to F\}$

FDs of F

$AB \to C$ ✓

$BC \to A$ ✓

$AC \to B$ ✓

$AD \to E$ ✓

$B \to D$

$(B)^{+}$ wrt $F_1 \cup F_2 \cup F_3 = \{B$

$\therefore B \to D$ is lost

$E \to F$

$(E)^{+}$ wrt $F_1 \cup F_2 \cup F_3 = \{F\}$

$\therefore E \to F$ is lost

D_1 is not a dep. Preserving decomposition

Q: Let R(ABCDEF) be the relational schema with FD set $F = \{AB \to C,\ BC \to A,\ AC \to B,\ B \to D,\ AD \to E,\ E \to F\}$

Which of the following decomposition of R is dependency preserving decomposition

$D_2 = \{R_1(ABC),\ R_2(ABDE),\ R_3(EF)\}$

$F_1 = \{AB \to C,\ AC \to B,\ BC \to A\}$

$F_2 = \{B \to D,\ AD \to E\}$

$F_3 = \{E \to F\}$

All FDs of F are implied by $F_1 \cup F_2 \cup F_3$

$\therefore F = F_1 \cup F_2 \cup F_3$

Hence D_2 is dep. preserving decomposition.

* H.W: Kindly go through the following relation algebra operation

1. Projection (π)
2. Selection (σ)
3. Cross product ($\times$)

Natural Join ($\bowtie$) is a derived relational algebra opn which is derived using

(i) Cross product
(ii) Selection (σ)
(iii) Projection (π)

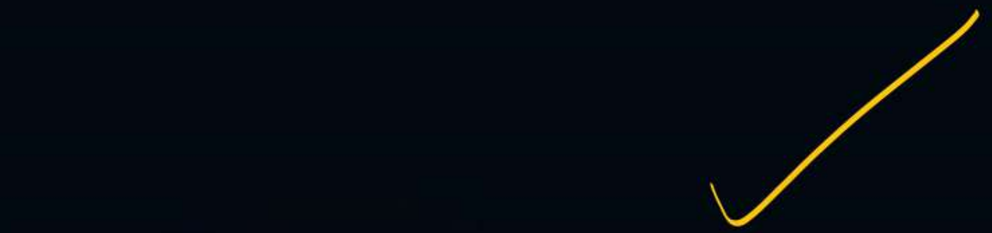


2 mins Summary

Topic Properties of decomposition

Topic Dependency preserving decomposition

Slide



THANK - YOU

Slide